

POLITECNICO
MILANO 1863

Best Bike Paths, Luca Suzani, Alberto Spezzapria, Alessio Pesavento

ITD Document

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Authors:	Luca Suzani, Alberto Spezzapria, Alessio Pesavento
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Introduction

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Implemented Requirements and Functions

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Testing

5.1 Testing

The implementation has undergone a rigorous testing phase focusing on the geospatial algorithms, data integrity constraints, user permission logic (Row Level Security), and privacy-aware workflows. The following subsections describe the main validation scenarios designed to ensure correctness, usability, and protection of community data.

5.1.1 Geospatial Path Recognition

This module verifies the system's ability to identify whether a newly recorded activity matches an existing path in the database using the PostGIS-powered `check_path_similarity` RPC function.

ID	Test Scenario	Expected Outcome	Result
GEO-01	**Perfect Match:** User records an activity that strictly follows the geometry of an existing public path within the 50m tolerance buffer.	The system identifies the existing path ID. The UI displays a "Match Found" dialog prompting the user to link the activity to the existing path instead of creating a new one.	**Pass**
GEO-02	**Partial Match / Deviation:** User records a track that overlaps with an existing path for 40% of the length but deviates significantly for the rest.	The system returns no match (null). The UI prompts the user to create a new path with a new name and description.	**Pass**
GEO-03	**Reverse Direction:** User traverses an existing path starting from the end point to the start point.	The algorithm recognizes the path regardless of the traversal direction.	**Pass**

Table 5.1: Geospatial Similarity Test Cases

5.1.2 Community Integrity and Data Protection

These tests validate the logic governed by the `get_path_community_usage` function and RLS policies, ensuring that public resources used by the community cannot be deleted or modified in ways that would compromise data integrity.

ID	Test Scenario	Expected Outcome	Result
SEC-01	**Safe Deletion:** An owner attempts to delete a path with 0 associated activities from other users.	The deletion proceeds successfully. The path and the owner's personal activities are removed (cascade delete).	**Pass**
SEC-02	**Protected Deletion:** An owner attempts to delete a public path that has been used by at least one other rider.	The database rejects the operation. The frontend displays a warning: "Cannot delete: Path is actively used by X riders".	**Pass**
SEC-03	**Privatization Attempt:** An owner switches a path from public to private while it has community activities.	The operation is blocked to preserve visibility and consistency of linked activities.	**Pass**

Table 5.2: Community Integrity Test Cases

5.1.3 Privacy and Workflow Consistency

This subsection validates the advanced workflow logic designed to prevent data duplication and respect user ownership priority during path matching and publication.

ID	Test Scenario	Expected Outcome	Result
WRK-01	**Ownership Priority:** User A has a private path P_{priv}. An identical public path P_{pub} also exists. User A records an activity on this route.	The system detects both paths but prioritizes ownership. The UI automatically links the activity to P_{priv} ("Welcome Back") without prompting to link to the public version.	**Pass**
WRK-02	**Tracking Opt-Out:** User records an activity on a public path but chooses not to appear on the leaderboard (unchecks "Link Activity").	The system creates a new **Private Copy** of the path for the user. The activity is linked to this new private entity, ensuring isolation from public stats.	**Pass**
WRK-03	**Publication Conflict:** User attempts to toggle a private path to "Public", but an identical public path already exists in the system.	The system blocks the direct status change to prevent duplicates. A dialog offers a choice: Merge into the existing path or cancel the operation.	**Pass**
WRK-04	**Merge Resolution:** User accepts the "Merge" option during a publication conflict.	The private path is deleted. All associated activities are transferred to the existing public path. The user receives a success confirmation.	**Pass**
WRK-05	**Passive Cancellation:** User chooses "Keep Private" during a publication conflict.	The publication process is aborted. The path remains private and no changes are made to the database (zero-side-effect action).	**Pass**

Table 5.3: Privacy and Workflow Logic Test Cases

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Chapter 6

Installation Instructions

Chapter 7

Effort Spent

Task	Date	Alessio P.	Alberto S.	Luca S.
App Implementation	10/01/2026	3h	2h	2h
	12/01/2026	2h	3h	2h
	15/01/2026	2h	2h	3h
	18/01/2026	1.5h	2h	2h
	Total	**8.5h**	**9h**	**9h**
Document Preparation	22/01/2026	2h	1.5h	1h
	24/01/2026	2h	2h	1.5h
	26/01/2026	1.5h	1.5h	2h
	Total	**5.5h**	**5h**	**4.5h**
TOTAL		**14h**	**14h**	**13.5h**

Table 7.1: Hours of work spent by each author on app implementation and document preparation